



to deliver believable tactile experiences in virtual environments. This new multisensory pseudo-haptic technology uses a combination of visual feedback from a VR headset

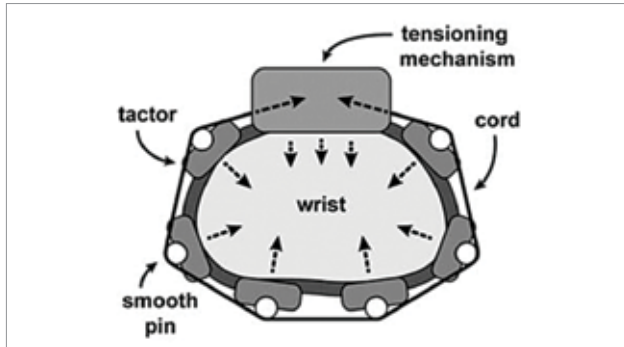


Illustration of the components in the Tasbi tactile bracelet (Credit: MAHI Lab/Rice University)

and tactile sensations from a mechanical bracelet that squeezes and vibrates the wrist. It includes both tactile sensations conveyed through skin and kinesthetic sensations from muscles and tendons. The device, named Tasbi, has a motorised cord that can tighten and squeeze the wrist, as well as a half-dozen small vibrating motors—the same components that deliver silent alerts on mobile phones—which are arrayed around the top, bottom, and sides of the wrist. When and how much these vibrate and when and how tightly the bracelet squeezes can be coordinated, both with one another and with a user's movements in virtual reality.

Issues being addressed to make solid-state batteries competitive

Researchers from Justus Liebig University Giessen, University of Münster, and Helmholtz Institute Münster of

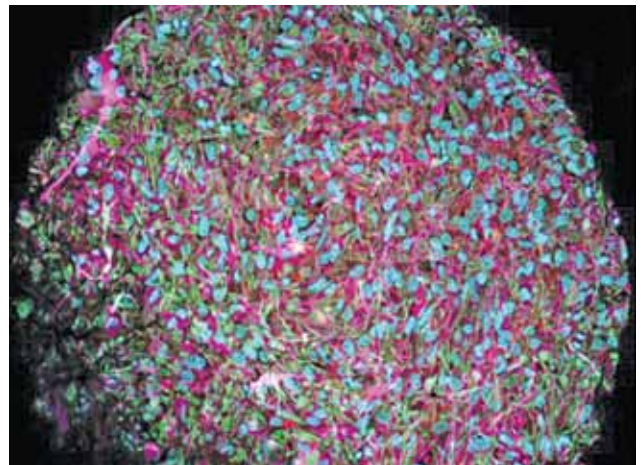


Doctoral student Laura Goodwin from Justus Liebig University Giessen with a solid-state cell on a laboratory scale (Credit: JLU/Rolf K. Wegst)

Forschungszentrum Jülich (Germany) analysed the current state of the technology, critically considering the challenges and the unresolved issues which must be addressed to make solid-state batteries competitive. The aim is to use a solid electrolyte in solid-state batteries, promising better storage properties, longer lifetimes, and increased safety. A key point for the development of solid electrolytes, which can ensure higher battery performance and safety at the same time, is having as low concentration of lithium as possible. Moreover, a maximum-capacity anode material is required, which enables the battery to have less volume and weight.

Brain-cells powered biocomputer could help modern computing

Researchers at Johns Hopkins University believe a biocomputer powered by human brain cells could be developed within our lifetime to exponentially expand the capabilities of modern computing and create novel fields of study. Thomas Hartung, a professor of environmental health sciences at Johns Hopkins, began to grow and assemble brain cells into functional organoids in 2012 using cells from human skin samples repro-



Magnified image of a brain organoid produced in Thomas Hartung's lab, dyed to show neurons in magenta, cell nuclei in blue, and other supporting cells in red and green (Credit: Jesse Plotkin/Johns Hopkins University)

grammed into an embryonic stem cell-like state. Each organoid contains about 50,000 cells, about the size of a fruit fly's nervous system. He now envisions building a futuristic computer with such brain organoids. But at current pace it would take decades before organoid intelligence can power a system as smart as a mouse. This problem can be tackled by scaling up production of brain organoids and training them with artificial intelligence.